01–teensy

File: sheets/01–teensy.kicad_sch

02–analog–front

File: sheets/02–analog–front.kicad_sch

03–adc

File: sheets/03–adc.kicad_sch

04–mcp–io

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05–cv–dac

File: sheets/05–cv–dac.kicad_sch

06–surface

File: sheets/06–surface.kicad_sch

07–midi

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08–connectors–panel

File: sheets/08–connectors–panel.kicad_sch

09–connectors–rear

File: sheets/09–connectors–rear.kicad_sch

10–power

File: sheets/10–power.kicad_sch

11–diagnostics

File: sheets/11–diagnostics.kicad_sch

12–i2c–branches

File: sheets/12–i2c–branches.kicad_sch

Sheet: /
File: pcb–m.kicad_sch

Title:

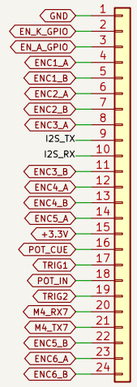
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 1/13

D45: Audio Shield Rev D en rieles propios junto a la Teensy; I2S por pistas cortas. Sin microSD ni pot del shield.

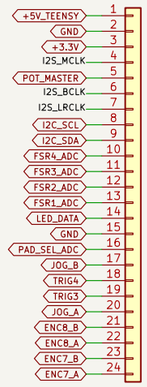
T25/T27/T38/T39 = TRIG1–4 e-drum (acondicionamiento DNP en 09). HP_VGND nunca se une a GND/AGND.

Filas de sockets conservadas; validar mecanica del montaje antes de fabricar.

J_T41L1
Teensy41-LeftRow



J_T41R1
Teensy41-RightRow



J_SHIELD_IN1
SHIELD pads LINE IN L/R/G



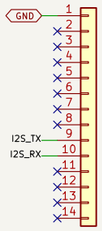
J_SHIELD_OUT1
SHIELD pads LINE OUT L/R/G



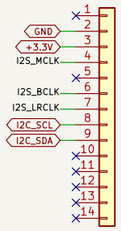
J_SHIELD_HP1
HP L/R/VGND — NEVER GND



J_SHD_L1
AudioShieldRevD-L



J_SHD_R1
AudioShieldRevD-R



Sheet: /01-teensy/
File: 01-teensy.kicad_sch

Title:

Size: A3 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 2/13

Frente analogico de la portadora Studio conservado (doc25 seccion 6).

D13: selector en panel: escalera 10k/10k a PAD_SEL_ADC. Polo C via J_PAD_SEL (3 hilos).

SW_PAD representa polos A/B en panel, cableados al header 2x4: no montar switch en PCB-M.

D17: R_ISO_IN_L/R 0 ohm 0805 separate front from codec: remove both if input fails.

R_PAD_L1
33k

R_PAD_L2
12k

R_PAD_L3
5.1k

SW_PAD1A
PAD 3P3T A/B

C_AC_L1
1uF

R_SER_L1
1k

R_BIAS_L1
100k

D_CLAMP_L1
BAT54S

JP_BYP_L1
BYP L (sin U_OA_IN)

U_OA_IN1A
MCP6004

TP_BUF_L1
BUF_L

C_IN_L1
4.7uF

R_PAD_R1
33k

R_PAD_R2
12k

R_PAD_R3
5.1k

SW_PAD1B
PAD 3P3T A/B

C_AC_R1
1uF

R_SER_R1
1k

R_BIAS_R1
100k

D_CLAMP_R1
BAT54S

JP_BYP_R1
BYP R (sin U_OA_IN)

U_OA_IN1B
MCP6004

TP_BUF_R1
BUF_R

C_IN_R1
4.7uF

R_VREF_T1
10k

R_VREF_B1
10k

C_VREF1
10uF

U_OA_IN1C
MCP6004

JP_BYP_VREF1
BYP VREF (sin U_OA_IN)

TP_VREF1
VREF

U_OA_IN1D
MCP6004

U_OA_IN1E
MCP6004

C_OA_IN1
100nF

C_LOUT_L1
10uF

R_LOUT_L1
47k

C_LOUT_R1
10uF

R_LOUT_R1
47k

R_SEL_T1
10k

R_SEL_B1
10k

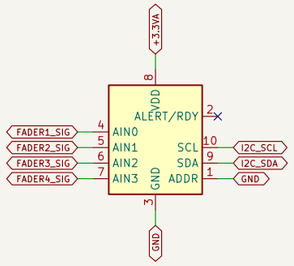
R_ISO_IN_L1
0

R_ISO_IN_R1
0

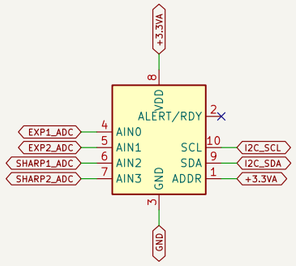
Sheet: /02-analog-front/		
File: 02-analog-front.kicad_sch		
Title:		
Size: A1	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 3/13

ADS1=0x48 (ADDR=GND); ADS2=0x49 (ADDR=VDD); ALERT/RDY no usado.

U_ADS1
ADS1115IDGSR



U_ADS2
ADS1115IDGSR



C_ADS1_VCC1
100nF



C_ADS2_VCC1
100nF



C_ADS1_BULK1
10uF



C_ADS2_BULK1
10uF



Sheet: /03-adc/
File: 03-adc.kicad_sch

Title:

Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 4/13

RESERVADO: MCP23017 0x23 vive en PCB-K. Hoja vacia conservada en la raiz.

Sheet: /04-mcp-io/	
File: 04-mcp-io.kicad_sch	
Title:	
Size: A3	Date:
KiCad E.D.A. 10.0.6	**Rev:** Id: 5/13

CV: 1 + 10k/20k = 1.5; A0 DAC1=GND (0x60), DAC2=3V3 (0x61).

D11: realimentacion antes de 1k serie: BAT54S en nodo protegido hacia jack, a GND/+5V_CV (D38).

D38: +5V_TEENSY -> SJ_CV + RC 10 ohm / 47uF -> +5V_CV; swing 4.95V ideal, medir margen/carga.

U_DAC1
MCP4725A0T-E/CH
+3.3V
I2C_SCL
I2C_SDA
GND
SCL
SDA
A0
VDD
VSS
VOUT
DAC1_RAW

U_OA_CV1A
MCP6004
DAC1_LIMITED
CV1_FB
CV1_OUT

R_CV_RF1
10k
CV1_OUT
CV1_FB

R_CV_RI1
20k
CV1_FB
GND

R_CV_SER1
1k
CV1_OUT
CV1_PROTECTED

D_CV_CLAMP1
BAT54S
GND
+5V_CV
CV1_PROTECTED

U_DAC2
MCP4725A0T-E/CH
+3.3V
I2C_SCL
I2C_SDA
GND
SCL
SDA
A0
VDD
VSS
VOUT
DAC2_RAW

U_OA_CV1B
MCP6004
DAC2_LIMITED
CV2_FB
CV2_OUT

R_CV_RF2
10k
CV2_OUT
CV2_FB

R_CV_RI2
20k
CV2_FB
GND

R_CV_SER2
1k
CV2_OUT
CV2_PROTECTED

D_CV_CLAMP2
BAT54S
GND
+5V_CV
CV2_PROTECTED

U_OA_CV1C
MCP6004
GND
CV_UNUSED_3
CV_UNUSED_3

U_OA_CV1D
MCP6004
GND
CV_UNUSED_4
CV_UNUSED_4

U_OA_CV1E
MCP6004
+5V_CV
V+
V-
GND

R_DAC_LIMIT1
2.2k
DAC1_RAW
DAC1_LIMITED

R_DAC_LIMIT2
2.2k
DAC2_RAW
DAC2_LIMITED

C_DAC1_VCC1
100nF
+3.3V
GND

C_DAC2_VCC1
100nF
+3.3V
GND

C_OA_CV_VCC1
100nF
+5V_CV
GND

R_CV_FILTER1
10
+5V_CV_IN
+5V_CV

C_DAC1_BULK1
10uF
+3.3V
GND

C_DAC2_BULK1
10uF
+3.3V
GND

C_OA_CV_BULK1
10uF
+5V_CV
GND

C_CV_FILTER1
47uF 16V
+5V_CV
GND

Sheet: /05-cv-dac/
File: 05-cv-dac.kicad_sch

Title:

Size: A2	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 6/13

Componentes de panel por cable. JOG_CLICK vía J_PCB_K hacia MCP 0x23.

D13: polo C común a J_PAD_SEL_2: extremos a 3,/_3: contacto central sin cable (1.65V).

D11: pots con RC 1k/100nF del lado ADC. J_HP único jack HP; HP_VGND aislado de masa.

J_FSR1
JST-PH_2pin_FSR1
+3.3V
FSR1_ADC
J_FSR2
JST-PH_2pin_FSR2
+3.3V
FSR2_ADC
J_FSR3
JST-PH_2pin_FSR3
+3.3V
FSR3_ADC
J_FSR4
JST-PH_2pin_FSR4
+3.3V
FSR4_ADC
R_FSR1
10k
FSR_ADC
GND
R_FSR2
10k
FSR_ADC
GND
R_FSR3
10k
FSR_ADC
GND
R_FSR4
10k
FSR_ADC
GND
J_JOG1
JOG_CABLE
GND
JOG_A
JOG_B
JOG_CLICK
J_POT_MASTER1
POT_MASTER_CABLE
+3.3V
POT_MASTER_WIPER
GND
J_POT_CUE1
POT_CUE_CABLE
+3.3V
POT_CUE_WIPER
GND
J_POT_IN1
POT_IN_CABLE
+3.3V
POT_IN_WIPER
GND
J_PAD_SEL1
PAD_SEL_CABLE
+3.3V
PAD_SEL_ADC
GND
J_HP1
HP_CABLE
HP_L
HP_R
HP_VGND
R_POT_MASTER1
1k
POT_MASTER_WIPER
POT_MASTER
R_POT_CUE1
1k
POT_CUE_WIPER
POT_CUE
R_POT_IN1
1k
POT_IN_WIPER
POT_IN
C_POT_MASTER1
100nF
POT_MASTER
GND
C_POT_CUE1
100nF
POT_CUE
GND
C_POT_IN1
100nF
POT_IN
GND
R_POT_BIAS_MASTER1
100k
POT_MASTER
GND
R_POT_BIAS_CUE1
100k
POT_CUE
GND
R_POT_BIAS_IN1
100k
POT_IN
GND

Sheet: /06-surface/
File: 06-surface.kicad_sch
Title:
Size: A1
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 7/13

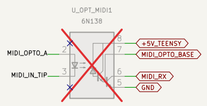
MIDI TRS Type A: ring=fuente, tip=sumidero; entrada aislada, sleeve IN NC.

D12 CA-033 (2014): R_MIDI_R=33 ohm >=0.5W; R_MIDI_T=10 ohm >=0.25W; sin 100 ohm extra.

JST-PH hacia jacks de panel: 1=tip, 2=ring, 3=sleeve. Sin deteccion de insercion.

D20: ALL DNP / excluded assembly BOM: USB MIDI on PI, no MIDI panel holes. Teensy 0=EN_K; 1=EN_A.





















Sheet: /07-midi/
File: 07-midi.kicad_sch

Title:

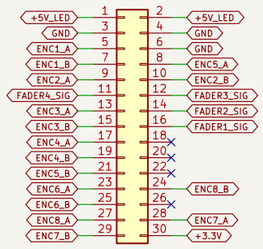
Size: A2	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 8/13

D8: J1 segun PCB-A fabricada: NC 18(spare)/20(INT_K)/22(INT_A)/26(LED_DIN).

Encoders a Teensy segun firmware; faders al ADS0x48 AIN0-3. LED solo via J_PCB_K.

I2C hacia PCB-A por J2: J1 no lleva SDA/SCL. Retirar pull-ups PCB-A.

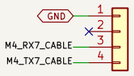
J1_PCBA1
IDC_2x15_PCB-A



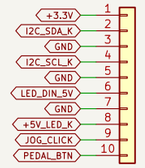
J2_I2C1
JST-SH_4pin_I2C



J3_M4
JST-PH_4pin_M4



J_PCB_K1
PCB_K_KEY_CABLE



J5_SHARP1
JST-PH_3pin_Sharp1



J6_SHARP2
JST-PH_3pin_Sharp2



J7_EXP1
JST-PH_2pin_EXP1



J8_EXP2
JST-PH_2pin_EXP2



C_SHARP1
10uF



C_SHARP2
10uF



R_EXP1
10k

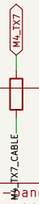
R_EXP2
10k

R_SHARP_LIMIT1
10k

R_SHARP_LIMIT2
10k

R_M4_RX1
1k

R_M4_TX1
1k



Sheet: /08-connectors-panel/
File: 08-connectors-panel.kicad_sch

Title:

Size: A3	Date:	**Rev:**
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DIS: Jacks con tuerca en pared trasera; JST-PH en borde PCB-M, uno por jack.

TRS: 1=tip/2=ring/3=sleeve. TS CV: 1=tip/2=sleeve. HP solo por J_HP (06).

MS: TVS unidirectional K=1 at PEDAL_BTN / A=2 at GND; cathode K marked on footprint.

Captura anterior sin contactos de detección: no se agrega detección de inserción.

Anillos LINE IN expuestos como redes locales: terminación balanceada sin definir.

J_LINE_IN_L1
LINE_IN_L_PANEL_CABLE
LINE_IN_L_T
LINE_IN_L_R
AGND

J_LINE_OUT_L1
LINE_OUT_L_PANEL_CABLE
LINE_OUT_L_T
AGND

J_CV1
CV1_PANEL_CABLE
CV1_PROTECTED
AGND

J_LINE_IN_R1
LINE_IN_R_PANEL_CABLE
LINE_IN_R_T
LINE_IN_R_R
AGND

J_LINE_OUT_R1
LINE_OUT_R_PANEL_CABLE
LINE_OUT_R_T
AGND

J_CV2
CV2_PANEL_CABLE
CV2_PROTECTED
AGND

J_PEDAL1
PEDAL_PANEL_CABLE
PEDAL_RAW
GND

J_EDRUM1
EDRUM_RESERVED_DNP
TRIG1_J
TRIG2_J
TRIG3_J
TRIG4_J
GND

R_PEDAL_SER1
1k
PEDAL_RAW
PEDAL_BTN

R_PEDAL_PU1
10k
+3.3V
PEDAL_BTN

C_PEDAL1
10nF
PEDAL_BTN
GND

D_PEDAL_ESD1
PESD5V3U1UL,315
PEDAL_BTN
GND

R_TRIG1_LOAD1
1M
TRIG1_J
GND

R_TRIG1_SER1
1k
TRIG1_J
TRIG1

D_TRIG1
BAT54S
GND
+3.3V
TRIG1

C_TRIG1
DNP
TRIG1
GND

R_TRIG2_LOAD1
1M
TRIG2_J
GND

R_TRIG2_SER1
1k
TRIG2_J
TRIG2

D_TRIG2
BAT54S
GND
+3.3V
TRIG2

C_TRIG2
DNP
TRIG2
GND

R_TRIG3_LOAD1
1M
TRIG3_J
GND

R_TRIG3_SER1
1k
TRIG3_J
TRIG3

D_TRIG3
BAT54S
GND
+3.3V
TRIG3

C_TRIG3
DNP
TRIG3
GND

R_TRIG4_LOAD1
1M
TRIG4_J
GND

R_TRIG4_SER1
1k
TRIG4_J
TRIG4

D_TRIG4
BAT54S
GND
+3.3V
TRIG4

C_TRIG4
DNP
TRIG4
GND

Sheet: /09-connectors-rear/
File: 09-connectors-rear.kicad_sch
Title:
Size: A1
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 10/13

LDO from +5V_TEENSY -> +3.3VA: +3.3V desde Teensy; +5V_LED separado de +5V_TEENSY. D10: NT1 junto a LDO, HP_VGND aislada.

D33: USB Pi -> F1 1812 -> PMOS D/S -> +5V_LED: SMAJ5.0A K1=rail/A2=GND; cathode K marked. Backup DNP.

D14: pull-ups 2.2k en PCB-M: desoldar R1/R2 de PCB-A. Unica union AGND/GND por NT1.

F1
0.75A MF-MSMF075/24-2

C_LDO_IN1
10uF

C_LDO_OUT1
1uF

Q_LED_REVERSE1
AO3401A

U_LDO1
TLV75533PDBV

U_LED_BUF1
74AHCT1G125

R_SDA1
2.2k

R_SCL1
2.2k

J_5V_BKUP1
5V_BACKUP_DNP

D_LED_TVS1
SMAJ5.0A

J_5V_LED_IN1
USB-A_PI_5V_LED_IN

NT1
Nettie_2

C_LED_BUF1
100nF

R_LED_BIAS1
100k

#FLG001
PWR_FLAG

#FLG002
PWR_FLAG

#FLG003
PWR_FLAG

#FLG004
PWR_FLAG

#FLG005
PWR_FLAG

#FLG006
PWR_FLAG

Sheet: /10-power/	
File: 10-power.kicad_sch	
Title:	
Size: A2	Date:
KiCad E.D.A. 10.0.6	**Rev:** Id: 11/13

DFT doc24 section 10. Bridges closed by copper: open copper then resolder to restore.

J1_PCB4 has no I2C: A branch is J2_I2C. Branch VCC isolation does not prevent signal backfeed.

Select one pull-up pair: 2.2k CLOSED default / 4.7k OPEN. Remove PCB-4 R1/R2; PCB-K pull-ups DNP.

TP_CV1/2 before 1k; TP_CV1/2_JACK after 1k. HP_XGND must remain separate.

R_OPT footprints DNP parallel only: MIDI reserved DNP, No change to MIDI OUT CA-033.

SJ_FRONT1
SJ_FRONT
+3.3VA
+3.3VA_FRONT

SJ_CV1
SJ_CV
+5V_TEENSY
+5V_CV_IN

SJ_LED_X1
SJ_LED_X
+5V_LED
+5V_LED_X

SJ_SHARP1
SJ_SHARP
+5V_LED
+5V_LED_SHARP

SJ_SDA_A1
SJ_SDA_A
I2C_SDA_B_A
I2C_SDA_A_SER

SJ_SCL_A1
SJ_SCL_A
I2C_SCL_B_A
I2C_SCL_A_SER

R_SDA_A7
4.7k
PU_SDA_A7
+3.3V

SJ_SDA_X1
SJ_SDA_X
I2C_SDA_B_X
I2C_SDA_X_SER

SJ_SCL_X1
SJ_SCL_X
I2C_SCL_B_X
I2C_SCL_X_SER

SJ_LED_DATA1
SJ_LED_DATA
LED_BUF_OUT
LED_DIN_5V

SJ_PU_SDA_22
SJ_PU_SDA_22
+3.3V
PU_SDA_22

SJ_PU_SDA_A7
SJ_PU_SDA_A7
+3.3V
PU_SDA_A7

R_SCL_A7
4.7k
PU_SCL_A7
+3.3V

R_IND_3V3
1.5k
+3.3V
IND_3V3_A

D_IND_3V3
RED-1mA
GND
IND_3V3_A

R_IND_3V3A1
1.5k
+3.3VA
IND_3V3A_A

SJ_PU_SCL_22
SJ_PU_SCL_22
+3.3V
PU_SCL_22

SJ_PU_SCL_A7
SJ_PU_SCL_A7
+3.3V
PU_SCL_A7

R_IND_5V_LED1
3.3k
+5V_LED
IND_5V_LED_A

D_IND_3V3A1
RED-1mA
AGND
IND_3V3A_A

D_IND_5V_LED1
RED-1mA
GND
IND_5V_LED_A

TP_5V_TEENSY1
+5V_TEENSY
+5V_TEENSY

TP_5V_RAW1
+5V_LED_RAW
+5V_LED_RAW

TP_5V_LED1
+5V_LED
+5V_LED

TP_3V3
+3.3V
+3.3V

TP_3V3A1
+3.3VA
+3.3VA

TP_FRONT1
+3.3VA_FRONT
+3.3VA_FRONT

TP_5V_CV1
+5V_CV
+5V_CV

TP_AGND1
AGND
AGND

TP_GND1
GND
GND

TP_SDA1
I2C_SDA
I2C_SDA

TP_SCL1
I2C_SCL
I2C_SCL

TP_LED_DIN1
LED_DIN_5V
LED_DIN_5V

TP_CV1
CV1_OUT
CV1_OUT

TP_CV2
CV2_OUT
CV2_OUT

TP_CV1_JACK1
CV1_PROTECTED
CV1_PROTECTED

R_OPT_BIAS_L1
DNP tune parallel
+3.3VA
HP_L

R_OPT_BIAS_R1
DNP tune parallel
+3.3VA
HP_R

R_OPT_MIDI_BIAS1
DNP tune parallel
+3.3V
MIDI_RX

R_OPT_MIDI_BASE1
DNP tune parallel
MIDI_RX_BASE
GND

TP_CV2_JACK1
CV2_PROTECTED
CV2_PROTECTED

#DFT001
PWR_FLAG
+3.3VA_FRONT

#DFT002
PWR_FLAG
+3.3V

#DFT003
PWR_FLAG
+5V_LED

#DFT004
PWR_FLAG
+5V_LED_RAW

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D21: TCA9517ADGK library graphic/pins = NXP PCA9517ADP C80799 (MSOP8); VCCA=VCCB=3.3V.

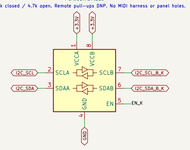
A=local bus shared: B_K and B_A separate cables. NEVER connect two B sides. EN changes only while idle.

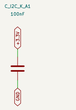
D30: EN_K=Teensy0, EN_A=Teensy1; separate 10k pull-ups / GPIO jumpers. Change EN only with bus idle.

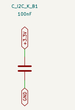
47 ohm 0805 series per cable line: replace with 0 ohm only after measuring VOL/rise time.

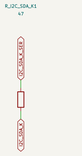
USBLC6-2SC6 C7519: 3.5pF max per line: short GND return to cable connector. PCB-A I2C only via J2.

Pull-ups per segment: 2.2k closed / 4.7k open. Remote pull-ups DNP. No MIDI harness or panel holes.















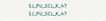








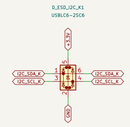


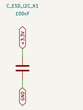


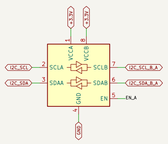


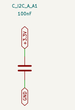
























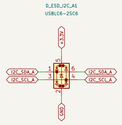


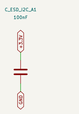


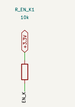




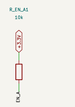


















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